# Rohan Malhotra

New York, NY | 973-713-3741 | ram9952@nyu.edu | rohanm.org | linkedin.com/in/rohanamal | github.com/rohanmalhotra0

## EDUCATION

**New York University, Courant Institute of Mathematical Sciences** — New York, NY
*B.A. in Mathematics and Computer Science (Double Major)* — *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Probability & Statistics, Real Analysis, Linear Algebra, Numerical Methods, Differential Equations

## EXPERIENCE

**Kalshi** — New York, NY
*Quantitative Developer* — *Jan 2026 – May 2026*

- Built a C++ hedging engine that converts contract prices (cents, probability, bps) into real-time exposure.
- Architected a WebSocket market-data pipeline ingesting quotes and trades to drive live pricing and hedge signals.
- Implemented fractional-Kelly position sizing with max-allocation caps to output recommended stake sizes.
- Co-authored research modeling job-loss risk as a macro-sensitive Poisson hazard with jump-diffusion income.
- Measured tail-risk reduction from Kalshi hedges with minimum-variance hedge ratios and Monte Carlo simulation.
- Built a Next.js/Python engine that recommends hedge size and compares cost vs. tail risk using FRED and BLS data.

**IBM** — Dallas, TX
*Robotics Engineer Intern* — *Jun 2026 – Aug 2026*

- Built a real-time perception stack for Boston Dynamics Spot with gRPC, multithreading, and lock-free queues.
- Trained YOLO11 models on 898 hand-labeled images to **99.5% mAP@50** with 99.4% precision and 99.4% recall.

*Applications Developer Intern, IBM Consulting* — *May 2026 – Aug 2026*

- Built an XGBoost cash-flow model, 13-day rolling cash forecasts, and SOFR loan models in Oracle EPM.
- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.

**Hume Center for National Security and Technology, Virginia Tech** — Blacksburg, VA
*Aerospace Researcher* — *Nov 2024 – May 2025*

- Programmed imaging and signal-processing tests in C for ContentCube, a CubeSat deployed from the ISS in 2025.
- Engineered a test environment simulating concurrent imaging workloads to resolve race conditions and memory bugs.

## RESEARCH

**Reddit Data in Quantitative Financial Models (published)** — Jan 2025 – May 2025
*Co-author, Virginia Tech (VTechWorks)* — *hdl.handle.net/10919/124730*

- Applied NLP sentiment scoring and regression models in Python to quantify sentiment's impact on equity returns.

**An Economic Approach to Optimize Capital Allocation (published)** — Aug 2024 – Dec 2024
*Author, Virginia Tech (VTechWorks)* — *hdl.handle.net/10919/124871*

- Ran Monte Carlo tests and backtests in Python comparing full vs. fractional Kelly under drawdowns and costs.

## PROJECTS

**Refrax** | refrax.io | *C++, Next.js, TypeScript, PostgreSQL* — Jun 2025 – Feb 2026

- Designed low-latency REST/WebSocket pipelines streaming market data into a PostgreSQL research environment.
- Implemented C++ risk engines (Greeks, P&L simulation, portfolio exposure) optimized for parallel execution.

**American Options Pricing System** | *Python, NumPy, Alpaca API*

- Built a scheduled pipeline that pulls option data, prices American options with a binomial tree, and logs runs.

## SKILLS

**Languages:** C++, Python, C, SQL, Java, Rust, R, MATLAB
**Quant & Math:** Stochastic processes, Monte Carlo, Kelly sizing, regression, time series, options pricing
**Libraries:** NumPy, Pandas, SciPy, statsmodels, scikit-learn, XGBoost, PyTorch
**Systems:** Linux, Git, Docker, PostgreSQL, WebSockets, gRPC, multithreading, lock-free queues, AWS